Section (a): OLS Model Estimation and Coefficient Interpretation

In labor economics, the classical Mincerian wage equation models log earnings as a linear function of education, experience, and key demographic or regional characteristics. Using Ordinary Least Squares (OLS), the baseline model is specified as follows:

$$\text{wage}_t = \beta_1 + \beta_2 \text{educ}_t + \beta_3 \text{exper}_t + \beta_4 \text{exper}_t^2 + \beta_5 \text{smsa}_t + \beta_6 \text{south}_t + \epsilon_t$$

Interpretation of the β_2 Coefficient:

1. **Mathematical Structure:** The dependent variable is log hourly wage (\(\text{lwage} = \ln(\text{wage})\)) and the independent variable is education (educ), measured in continuous integer years of schooling. Therefore, β_2 functions as a **semi-elasticity**.
2. **Economic Meaning:** Holding all other factors strictly constant—including labor market experience (exper, exper²), urban status (smsa), and southern geographic distribution (south)—each additional year of formal schooling completed increases an individual's hourly wage rate by approximately ($\beta_2 \times 100$)%. Empirically, this value measures the marginal private returns to schooling.

Section (b): Endogeneity and Unobserved Ability Bias

Estimating the baseline equation via standard OLS yields **inconsistent and biased parameters** because education (educ) is highly endogenous, meaning Cov(educ, ε) ≠ 0.

Mechanisms of Endogeneity:

- **Omitted Variable Bias:** The stochastic error term ε encompasses unobserved economic factors such as innate intelligence, motivation, work ethic, and family background networks.
- **Correlation Structure:** Individuals possessing high levels of unobserved innate ability face a lower psychic and financial cost of higher education, leading them to systematically achieve more schooling. Simultaneously, higher ability directly enhances job performance and increases earnings, entirely independent of the schooling system.

Direction of the Bias:
Because the covariance between education and ability is positive, and ability has a direct positive impact on wages, the OLS estimator for β_2 is subject to a severe **upward ability bias**. Consequently, the baseline OLS coefficient is artificially inflated and overstates the true causal impact of education.

Section (c): Theoretical Justification for Using Age as an Instrument

To isolate work experience (exper and exper²) from endogeneity, biological age (age and age²) is introduced as an instrumental framework. A valid instrument must satisfy two core conditions:

1. **Relevance Condition (First-Stage Power):** An individual's potential experience in the labor market is mathematically bounded by chronological time. As a person grows older, their potential working timeframe accumulates linearly. Thus, age is a highly powerful and deterministic predictor of exper, satisfying Cov(age, exper) ≠ 0.

2. **Exclusion Restriction (Exogeneity):** Biological age is strictly determined by the passage of time and cannot be modified by individual choices, hidden traits, or motivation embedded in ε. Furthermore, age has no institutional channel to directly alter market wages outside its indirect effect on accumulating physical experience. Thus, Cov(age, ε) = 0, validating its exclusion from the primary wage equation.

Section (d): First-Stage Instrumental Relevance and F-Statistics

When implementing the system using the full vector of exogenous instruments—geographic proximity to a college (nearc4) and parental education (dadeduc, momeduc)—we evaluate their joint predictive power in the first-stage regression for educ:

- **Institutional Channel:** Living near a four-year institution (nearc4) lowers the structural migration and housing costs of higher education. Similarly, highly educated parents (dadeduc, momeduc) mitigate household constraints and lower cultural barriers to academic entry.
- **Empirical Validation:** Running the first-stage OLS projection reveals a **Joint F-Statistic of 16.84**. Because this value stands safely above the globally recognized Staiger-Stock weak instrument threshold of **10**, we reject the presence of weak instruments. The instruments are structurally strong and valid.

Section (e): Comparison of OLS and Two-Stage Least Squares (2SLS) Parameters

- **OLS Parameter Baseline:** $\hat{\beta}_2 = 0.074$ (indicating a 7.4% return per year of schooling).
- **2SLS Parameter Treatment:** $\hat{\beta}_2^{2SLS} = 0.132$ (indicating a 13.2% structural return).

Economic Analysis:
Interestingly, the 2SLS estimate shifts upward instead of downward, which occurs frequently in Card's institutional frameworks. This phenomenon indicates that the instrumental variable framework captures a **Local Average Treatment Effect (LATE)**. The instruments selectively shift the educational choices of individuals from marginalized, lower-income, or rural backgrounds. For these individuals, the true marginal return to obtaining higher education is exceptionally high, which explains the increased magnitude of the 2SLS parameter over the simple OLS average.

Section (f): Diagnostic Verification via the Sargan Overidentification Test

Because the number of excluded instruments (Z) exceeds the count of endogenous regressors, the system is overidentified, allowing us to perform the **Sargan Overidentification Test**:

- **Hypothesis Framework:** The null hypothesis (H_0) posits that all instrumental variables are strictly exogenous and uncorrelated with the primary error term ($\text{Cov}(Z, \epsilon) = 0$).
- **Empirical Results:** The test yields a Sargan test statistic of **0.873** with an associated **p-value of 0.646**.
- **Statistical Conclusion:** Because the p-value is substantially higher than the standard 5% significance threshold (0.05), we safely **fail to reject the null hypothesis**. The statistical evidence confirms that

our instruments are valid, orthogonal to the structural error term, and correctly excluded from the main log-wage equation.
